

proof shows why a claim follows from accepted starting points.

FOUNDATIONS

A

A **proposition** is a statement that is true or false. Letters p or q stand for statements.

LOGIC SYMBOLS AND STATEMENTS



In everyday examples, treat each "if ... then ..." sentence as an assumed premise.

RULES OF INFERENCE



Modus Ponens or Rules of Inference

p

Simplification

RULES OF INFERENCE

p

Modus Tollens

RULES OF INFERENCE

p

Hypothetical syllogism RULES OF INFERENCE

p

Disjunctive Syllogism

RULES OF INFERENCE

p

BUILDING A PROOF WITH I

BUILDING A PROOF WITH INFERENCE RULES

Compare each proposed conclusion with what the premises actually allow.

FALLACIES

V

UNDERSTANDING AN IMPLICATION

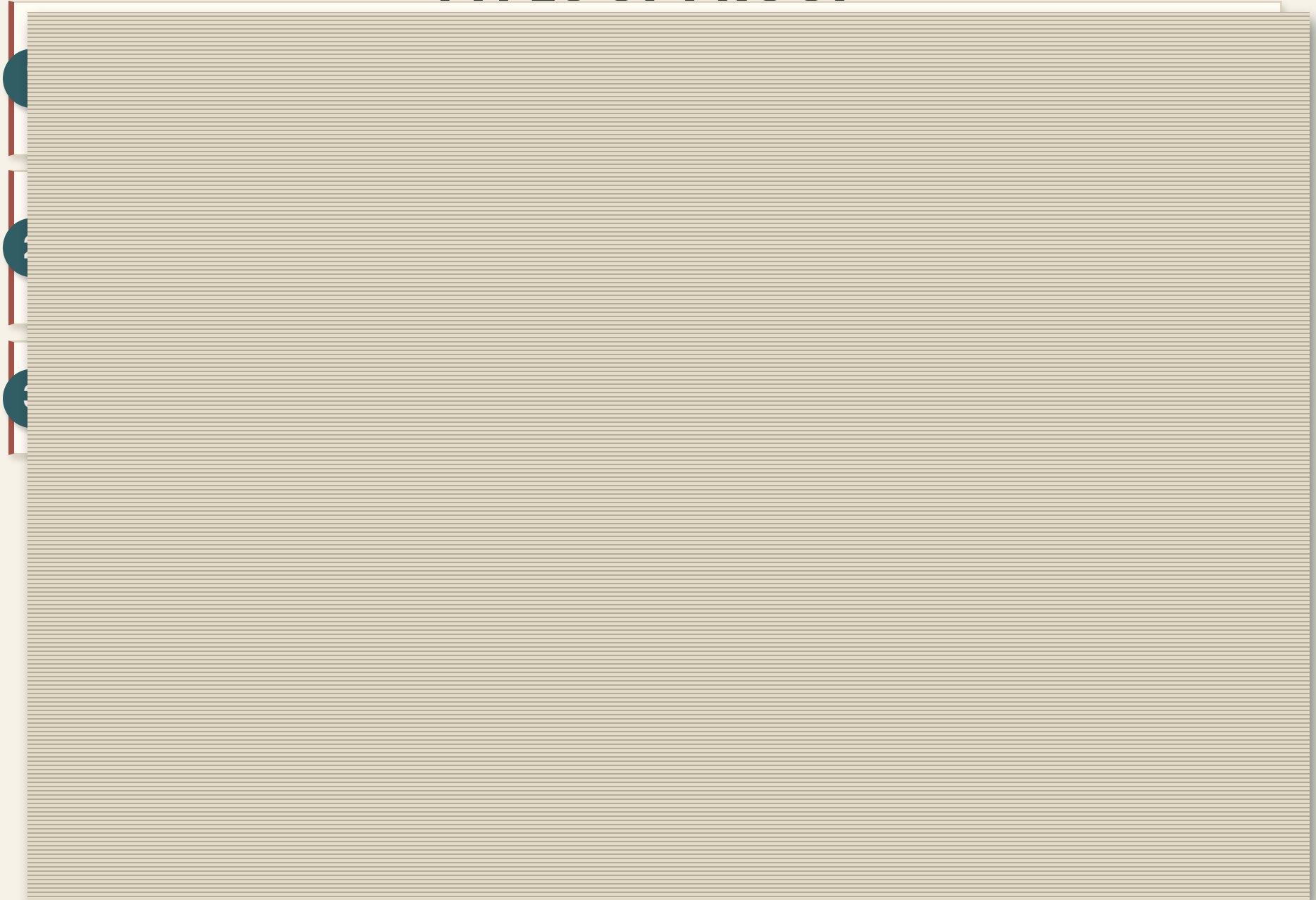
How do you verify the d. Here

Try all four cases

c
T
is
W
n
q
A
p
c

Three introductory approaches over the integers:

TYPES OF PROOF



FOR EVERY INTEGER x , IF 6 DIVIDES x , THEN 3 DIVIDES x .

G
a

GUIDED PROOF PRACTICE

For every integer n , if 4 divides n , then 2 divides n .

Hint: Consider the definition of divisibility. Can you find a counterexample?

So

Try

CHALLENGES

Given p and q . Which rule gives q ?

⊃ Modus Ponens ⊃ Modus Tollens ⊃ Simplification

State the claim and identify its assumptions.

THE FINAL LINE

THANK YOU

...

Q

A